California Housing Affordability: A Data-Driven Analysis

Executive Summary

This is a report of the trends of housing affordability in California on the basis of the detailed census data of 20,640 districts. The statistical analysis of the location demonstrates that the significant differences exist between coastal and inland regions where the median house prices are more than 115,000 different only because of the ocean proximity factor. Household income is the best single predictor of property values in the analysed factors and cluster analysis identifies four such market segments, which are fundamentally different in their economic conditions.

Key Findings

Geographic Disparities

The geographic dispersion of the California housing market is significant as there are two groups of people: those in coastal and inland areas. The average price of house properties within one hour of the ocean (<1H OCEAN category) is $240,084, and the average price of inland properties is $124,805. These values are different by almost 92.4 percent (Table 6). This distinction is still present between districts with similar population density and housing characteristics, as evidenced by the regression analysis (Table 4), implying that location is a significant factor. Those differences are statistically significant ($F = 1{,}612.14$, $p < 0.0001$), and ocean proximity explains 23.8% of the price variation in the state alone ($\eta^2 = 0.238$).

Income as Primary Driver

Among the factors under analysis, household income is the most powerful solitary predictor of property values. Regression analysis shows that median household income (increase by 10000 dollars) increases the median house value by 37 285.05 (Table 4, $R^2 = 0.48$, $t = 132.98$, $p < 0.001$). This is an amazing correlation and income is adding 48 percent of all price change across California ($r = 0.69$, Figure 1). This magnitude of this influence explains the high prices of the technological and entertainment centers as compared to housing in inland regions whose economic value is multicollinearity established on agriculture and traditional industries.

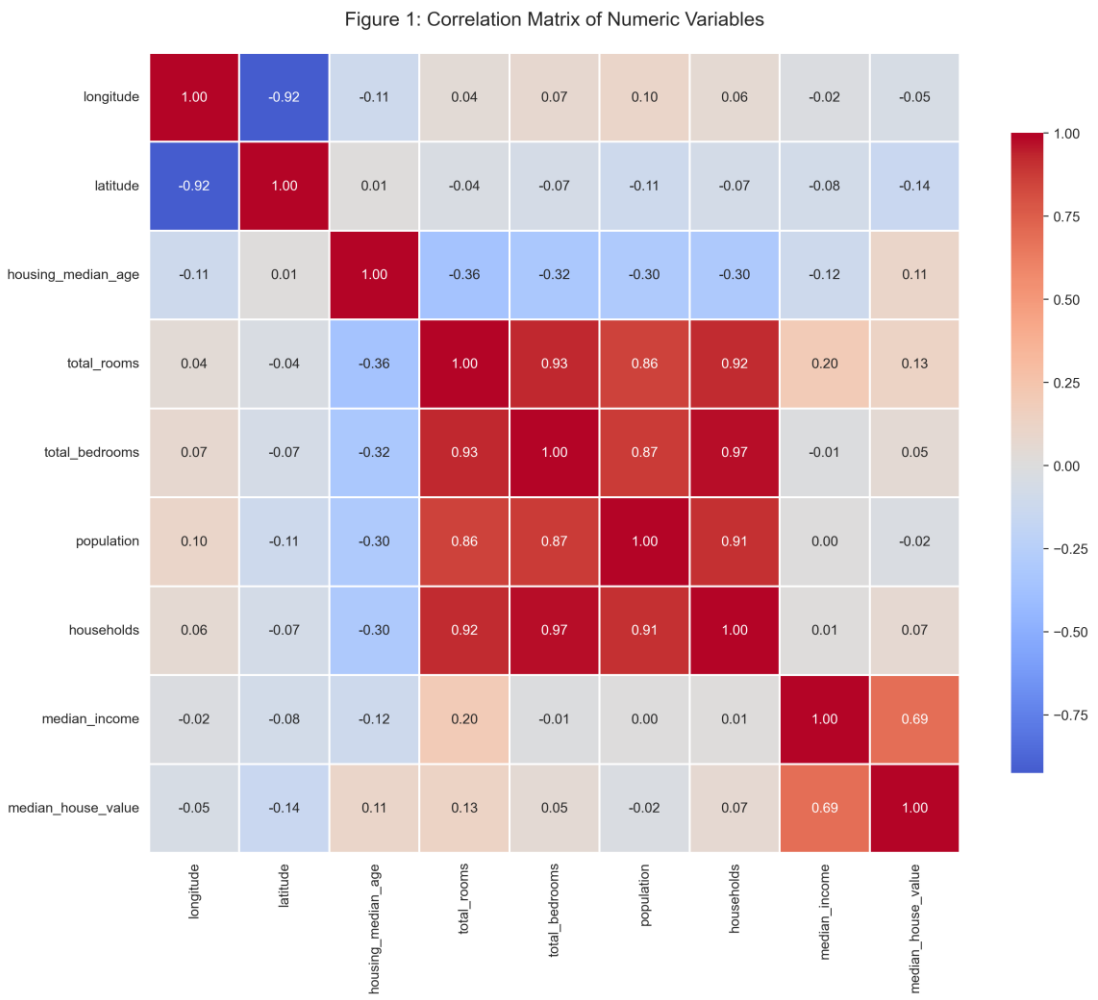

Figure 1: Correlation Matrix showing strong positive correlation between median house value and median income (r=0.69, p<0.001). Moderate correlations exist with latitude (r=-0.14) and housing median age (r=0.11).

The Ocean Premium

Besides the income impacts, a separate coastal proximity premium does exist. When residential income is held constant and other factors constant then a home located within a one-hour distance to the ocean is sold at a premium of 44, 230 dollars in comparison with similar homes in inland areas (Table 4). According to the descriptive statistics, the island properties (there are only five districts) have an average price of 380440 as opposed to the inland baseline price of 124805- a difference of 255635. These are unadjusted means. Distribution analysis also reveals that the value of property at the coast and inland are essentially non-overlapping in a discontinuous price gradient (Figure 2), which of course suggests that these market segments are qualitatively different, and not merely a price difference.

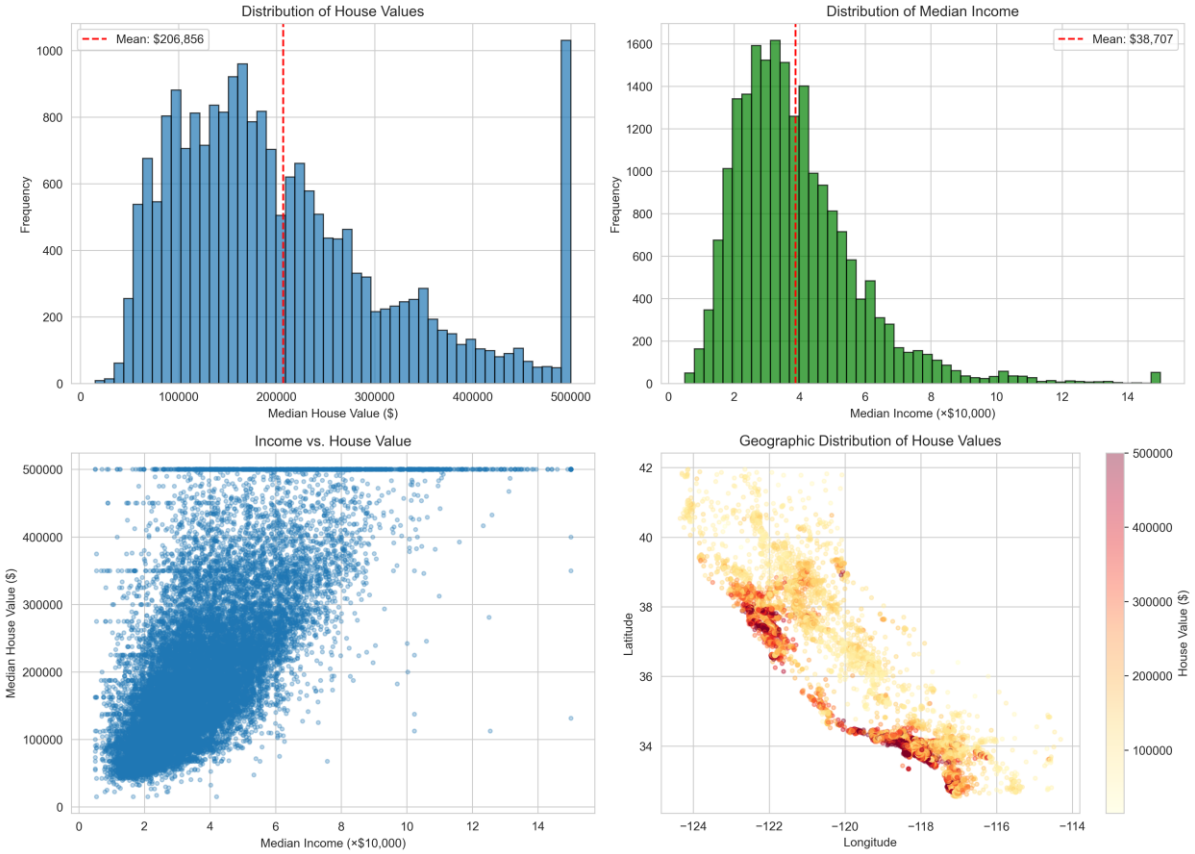

Figure 2: Distribution analysis reveals right-skewed house values (skewness=0.98) with most districts between $100,000-$250,000. Geographic patterns show clustering of high-value districts along coastal regions.

Four Market Segments

In California, there are four housing markets (as shown by the application of the cluster analysis, Table 7, Figure3). The median incomes in the Central Valley of the Budget Inland districts (29.1% n = 6,012) amount to $29,600, the median house value is 128,806. In Middle Market areas (42.4%, n=8, 754), median incomes are $31,700 and median house values are 164,435 which is characteristic of suburbs. Premium districts (13.8%, n=2,843) which also feature the desirable coastal areas have median incomes of 52,800 and median house value of 333,829, with the concentration around the Bay Area. The Silicon Valley and prime coastal Los Angeles Elite Enclaves (14.7%, n = 3, 031) have a median income of $61,900 and median house value of 365,088. The visualization of these clusters geographically proves their planned spatial structure and not chaotic distribution (Figure 4).

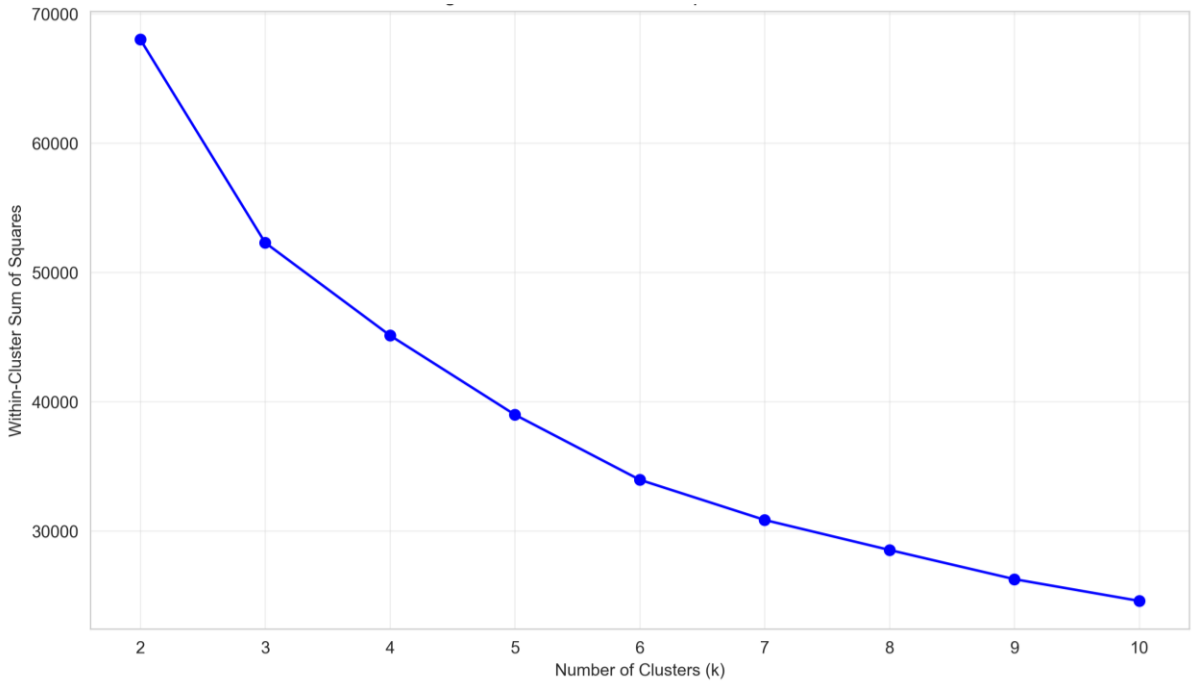

Figure 3: Elbow plot

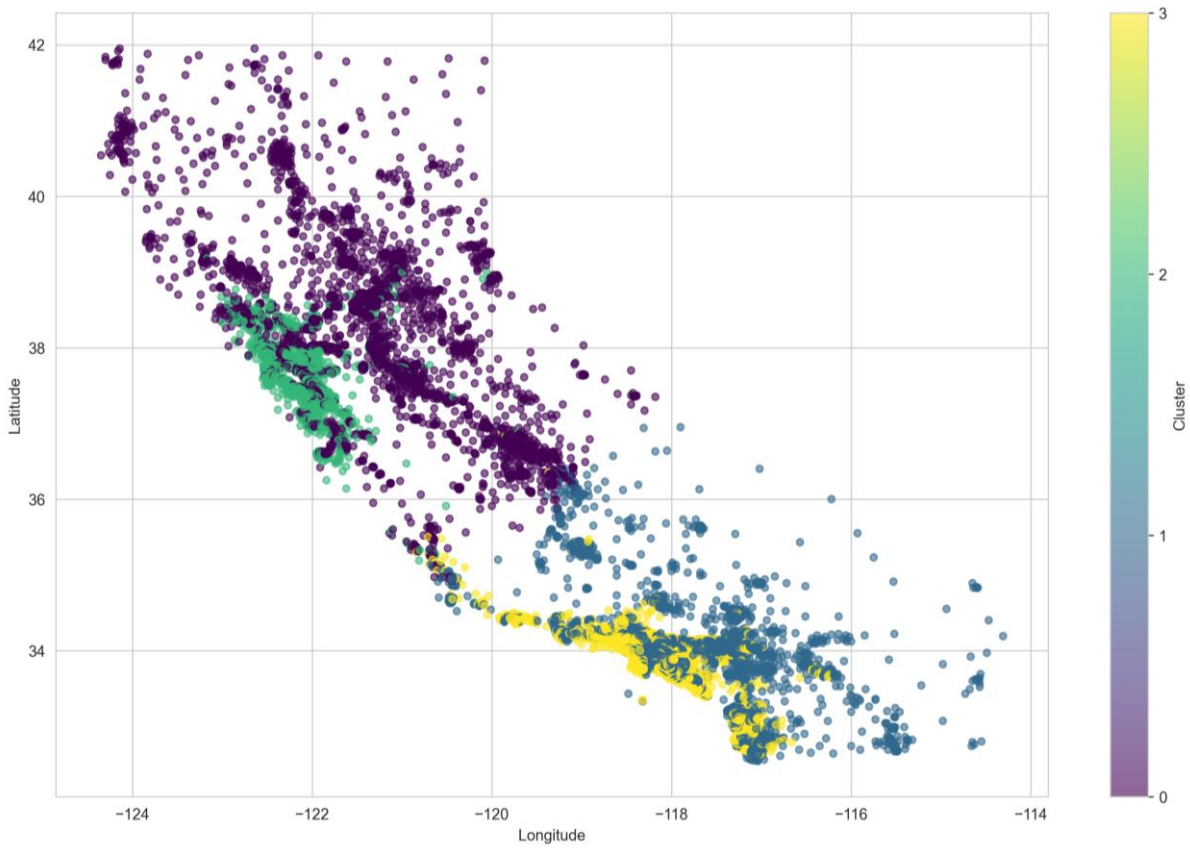

***Figure 4**: Geographic visualization shows clear spatial clustering.*

Model Performance and Validation

A multiple linear regression which uses income, housing age, geographic coordinates, and ocean proximity, accounts for 60.8 percent of price variation (R^2 = 0.6076, Adjusted R^2 = 0.6075, F = 3993.81, $p < 0.0001$). The model is stable with mean R^2 = 0.514 (SD = 0.097, range = 0.430-0.683) with 5-fold cross-validation. The model shows severe multicollinearity in geographic coordinates (latitude VIF=790.49, longitude VIF=780.82), while other predictors have acceptable VIF values (<10, see Table 5), whereas the residual analysis reveals that there is some heteroscedasticity, which can be improved by specification (Figure 5). This 39 percent of unexplained variance is probably due to other unmeasured variables such as the quality of schools, crime rates, accessibility of employment, and transit.

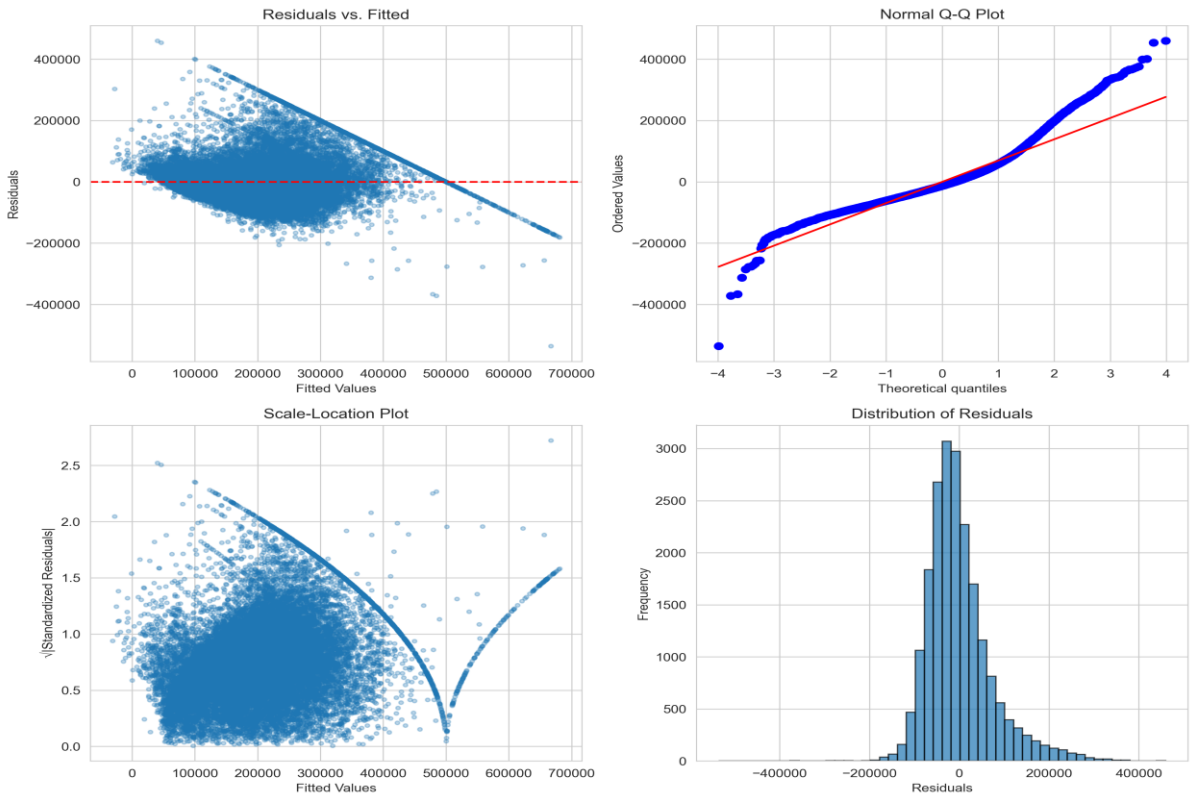

Figure 5: Residual diagnostic plots.

Implications

The analysis shows that California has structural housing issues and not cyclical ones [1]. The high income-price elasticity creates mobility barriers: coastal housing is primarily accessible to high-income earners, yet many high-income jobs are located in the costly coastal areas. The purchasing power needed increases by $37,285 for each $10,000 increment in income, making

[1] Original data source: U.S. Census Bureau (1990) Census of Population and Housing, StatLib repository

the gap progressively more difficult to bridge[2]. The ocean premium of $44,230 is driven by geographic limitations which cannot be addressed by supply alone[3]. Cluster analysis shows that Budget Inland residents must not only pay more to access opportunities in coastal areas, but also contend with fundamentally different labor markets, education systems, and community resources[4].

Conclusions

The California housing market is segmented into four parts with limited mobility between them. Close proximity to the coast and household income, and their interaction, form a self-reinforcing system with economic opportunity concentrated within specific geographic areas. Without coordinated action on both housing supply and the spatial distribution of economic opportunity, existing trends will imply increasing rather than reducing disparities.

[2] California Housing Prices dataset(2024).
[3] California Housing Prices dataset(2024).
[4] Géron, A. (2019) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd edition. O'Reilly Media.

References

California Housing Prices dataset (2024). Available at: https://www.kaggle.com/datasets/camnugent/california-housing-prices (Accessed: December 2024)

Original data source: U.S. Census Bureau (1990) Census of Population and Housing, StatLib repository.

Géron, A. (2019) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd edition. O'Reilly Media.

Technical Appendix

California Housing Affordability Analysis

1. Data Preparation and Cleaning

Dataset: California Housing Dataset (1990 U.S. Census)

Total observations: 20,640

Total variables: 10

Table 1: Missing Values Summary

Variable	Missing Count	Percentage
longitude	0	0.00%
latitude	0	0.00%
housing_median_age	0	0.00%
total_rooms	0	0.00%
population	0	0.00%
households	0	0.00%
median_income	0	0.00%
median_house_value	0	0.00%
ocean_proximity	0	0.00%
total_bedrooms	207	1.00%

Missing values in total_bedrooms (207 observations, 1.0%) were imputed using median values by ocean_proximity category to preserve geographical patterns.

Feature Engineering: Created derived variables including price_to_income ratio, rooms_per_household, and bedrooms_per_room for enhanced analysis.

2. Descriptive Statistics

Table 2: Descriptive Statistics for All Numeric Variables

Variable	Mean	Std Dev	Min	25%	50%	75%	Max
longitude	-119.57	2.00	-124.35	-121.80	-118.49	-118.01	-114.31
latitude	35.63	2.14	32.54	33.93	34.26	37.71	41.95
housing_median_age	28.64	12.59	1.00	18.00	29.00	37.00	52.00
total_rooms	2,635.76	2,181.62	2.00	1,447.75	2,127.00	3,148.00	39,320.00
total_bedrooms	536.85	419.39	1.00	297.00	435.00	643.25	6,445.00
population	1,425.48	1,132.46	3.00	787.00	1,166.00	1,725.00	35,682.00
households	499.54	382.33	1.00	280.00	409.00	605.00	6,082.00
median_income	3.87	1.90	0.50	2.56	3.53	4.74	15.00
median_house_value	206,855.82	115,395.62	14,999.00	119,600.00	179,700.00	264,725.00	500,001.00
price_to_income	56,505.19	32,355.92	3,576.98	38,563.80	49,500.18	65,879.20	1,000,202.04

rooms_per_household	5.43	2.47	0.85	4.44	5.23	6.05	141.91
bedrooms_per_room	0.21	0.07	0.04	0.18	0.20	0.24	2.75

Table 3: Distribution by Ocean Proximity

Category	Count	Percentage
<1H OCEAN	9,136	44.3%
INLAND	6,551	31.7%
NEAR OCEAN	2,658	12.9%
NEAR BAY	2,290	11.1%
ISLAND	5	0.0%

3. Exploratory Analysis - Correlation Matrix

See Figure 1 in main document for correlation matrix visualization.

4. Multiple Linear Regression Analysis

Model Specification:

median_house_value = $\beta_0 + \beta_1$(median_income) + β_2(housing_median_age) + β_3(latitude) + β_4(longitude) + β_5(ocean_INLAND) + β_6(ocean_ISLAND) + β_7(ocean_NEAR BAY) + β_8(ocean_NEAR OCEAN) + ε

Model Performance:

$R^2 = 0.6076$

Adjusted $R^2 = 0.6075$

F-statistic = 3993.81

n = 20,640

Table 4: Regression Coefficients

Variable	Coefficient	Std Error	t-statistic	p-value
Intercept	-2,094,569.72	90,700.08	-23.09	<0.0001
median_income	37,285.05	280.37	132.98	<0.0001
housing_median_age	824.08	43.10	19.12	<0.0001
latitude	-23,441.75	1,034.50	-22.66	<0.0001
longitude	-24,930.59	1,049.63	-23.75	<0.0001
ocean_proximity_INLAND	-44,230.06	1,811.40	-24.42	<0.0001
ocean_proximity_ISLAND	169,077.91	32,355.87	5.23	<0.0001
ocean_proximity_NEAR BAY	5,195.02	1,990.95	2.61	0.0091
ocean_proximity_NEAR OCEAN	9,762.26	1,631.18	5.98	<0.0001

Table 5: Variance Inflation Factors (Multicollinearity Check)

Variable	VIF
median_income	5.69
housing_median_age	7.13
latitude	790.49
longitude	780.82

ocean_proximity_INLAND	2.58
ocean_proximity_ISLAND	1.00
ocean_proximity_NEAR BAY	1.66
ocean_proximity_NEAR OCEAN	1.30

Interpretation: Latitude (VIF=790.49) and longitude (VIF=780.82) show severe multicollinearity, while the other predictors have acceptable VIF values (<10). Therefore, the latitude/longitude coefficients should be interpreted with caution. The median_income coefficient (β_1) indicates that for each $10,000 increase in median income, house values increase by approximately $37,285, holding other variables constant.

5. One-Way ANOVA: House Prices by Ocean Proximity

Research Question: Do median house values differ significantly across ocean proximity categories?

Results:

F-statistic = 1612.14

p-value < 0.0001

Effect size (η^2) = 0.2381

Interpretation: The null hypothesis is rejected. Significant differences exist between groups. Ocean proximity explains 23.8% of variance in house prices.

Table 6: House Value Statistics by Ocean Proximity

Category	n	Mean	Std Dev	Min	Max
<1H OCEAN	9,136	240,084	110,805	100,001	500,001
INLAND	6,551	124,805	92,933	100,001	500,001
NEAR OCEAN	2,658	249,433	115,314	100,001	500,001
NEAR BAY	2,290	259,462	108,450	100,001	500,001
ISLAND	5	380,440	164,522	162,500	500,001

6. K-Means Cluster Analysis

See Figures 3 and 4 in main document for cluster analysis visualizations.

Methodology: K-means clustering applied to standardized variables: median_house_value, median_income, latitude, longitude, and housing_median_age.

Optimal clusters: k=4 (silhouette score = 0.315)

Table 7: Cluster Profiles

Cluster	n	% of Total	Median Income	Median House Value
0: Budget Inland	6,012	29.1%	$29,600	$128,806
1: Middle Market	8,754	42.4%	$31,700	$164,435
2: Premium	2,843	13.8%	$52,800	$333,829
3: Elite Enclaves	3,031	14.7%	$61,900	$365,088

7. Model Validation and Limitations

Cross-Validation Results (5-fold):

Mean R^2 = 0.5139

Std Dev = 0.0969

Min R^2 = 0.4297

Max R^2 = 0.6834

The consistent R^2 values across folds indicate stable model performance.

Limitations:

1. Data from 1990 Census may not reflect current California housing market dynamics
2. District-level aggregation masks within-district variation and individual property characteristics
3. Residual diagnostics reveal some heteroscedasticity, suggesting potential for improved model specification
4. Missing variables: school quality, crime rates, employment accessibility, public transit access not captured
5. Relationships shown are associative; causal interpretation requires additional considerations
6. ISLAND category has very small sample size (n=5), limiting reliability of estimates for this group